Cerebellar-Inspired Neural Architectures for Motor Learning in Simulated Environments

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Introduction

This thesis will be based on two papers:

- Cerebellar-driven cortical dynamics can enable task acquisition, switching and consolidation
- Policy-Gradient Reinforcement Learning as a General Theory of Practice-Based Motor Skill Learning

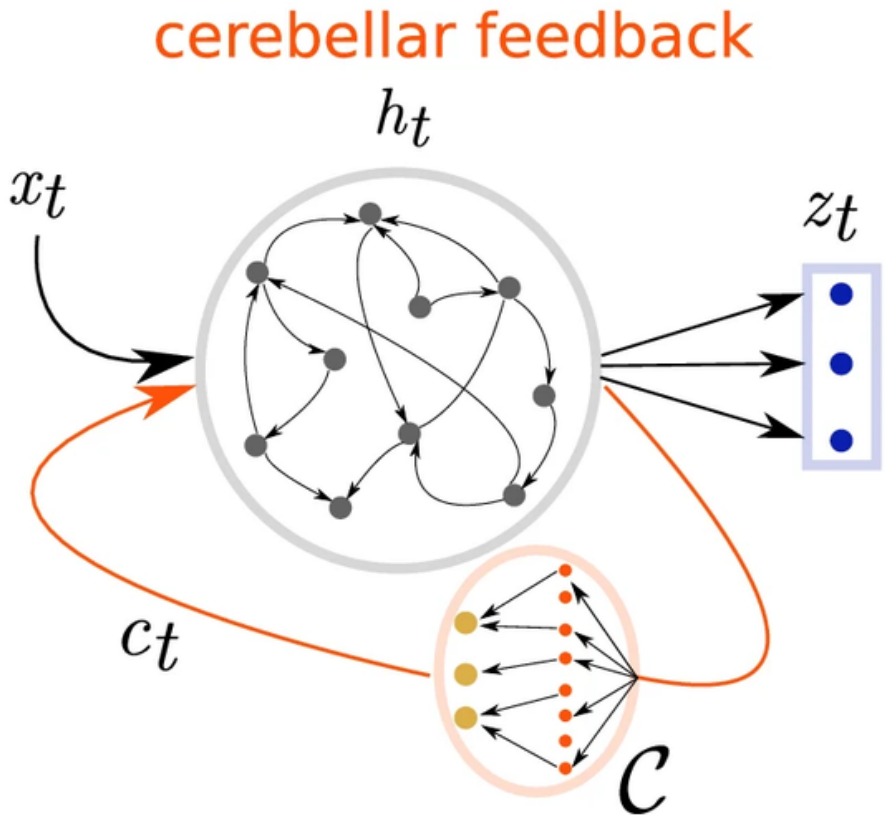
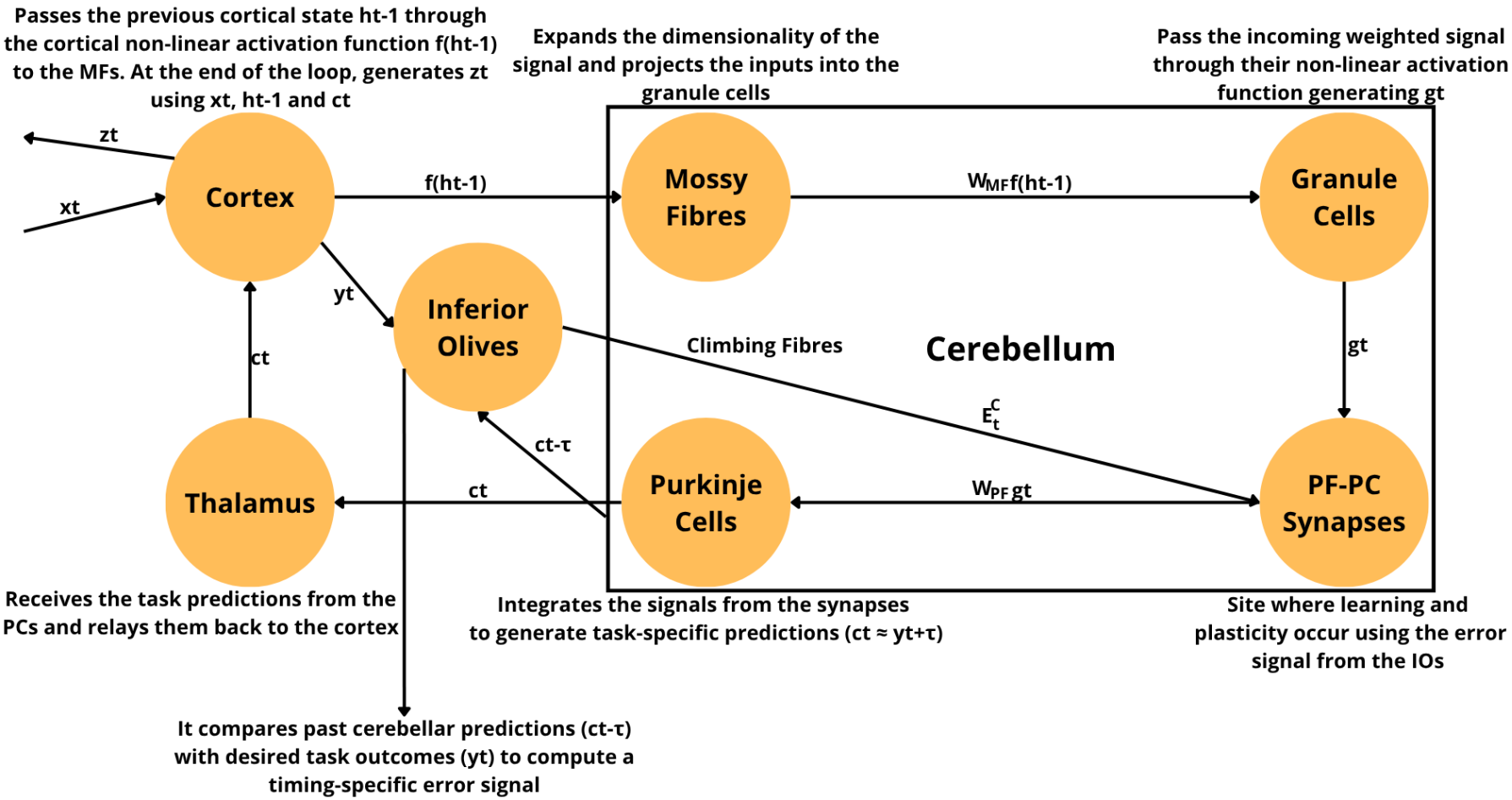


Figure: Cerebro-cerebellar network architecture.

Modeled Cerebellar Feedback Loop



Model Computational Representation

- Cerebellar Feedback Model

$$g_t = f^C(W_{MF} f(h_{t-1})) \tag{1}$$

$$c_t = C(f(h_{t-1})) = W_{PF} g_t \tag{2}$$

$$h_t = \alpha h_{t-1} + W_{hh} f(h_{t-1}) + W_{ih} x_t + W_{Ch} c_t \tag{3}$$

- Final Behavior Output

$$z_t = W_{rdt} f(h_t) \tag{4}$$

Model Training Error Functions

Two types of error functions are used:

- **Task Error:** Biologically plausible gradient descent is used to train the weights in h_t. The temporal error function to be minimized is:

$$E_t = \varepsilon(z_t, y_t) \tag{5}$$

  where y_t denotes the desired task outcome at time t and ε is the task error function (MSE for regression, Cross-Entropy loss for classification).

- **Cerebellar Error:** Internal learning rule used exclusively to optimize the cerebellum (c_t). It compares a past cerebellar prediction with the current desired outcome:

$$E_t^c = \varepsilon(c_{t-\tau}, y_t) \tag{6}$$

  where τ is the cerebellar learning time window (delay).

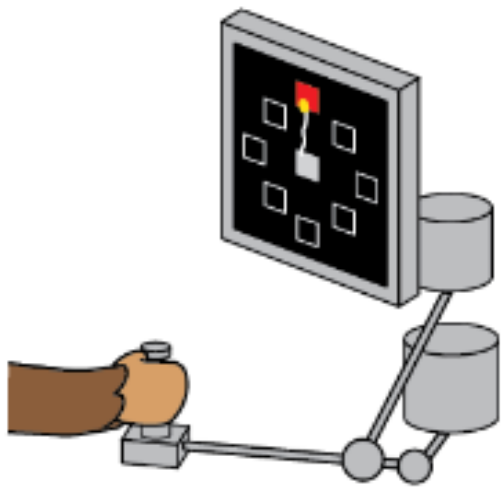
Cerebellar Feedback Model

Three types of cerebellar feedback models are considered:

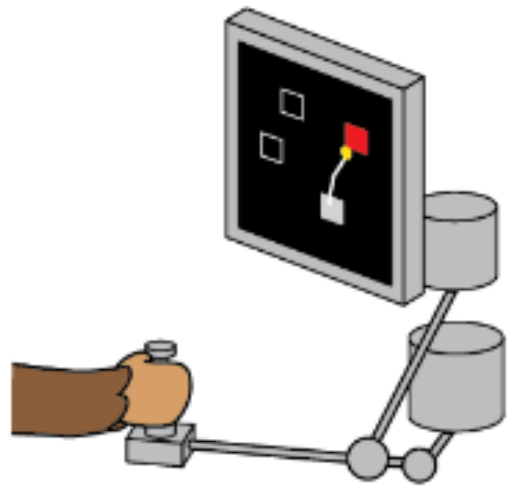
- **Fixed RNN:** The internal connectivity of the cortical RNN is fixed and random, and only the input weights are trained. (W_{hh}, W_{ih}, W_{MF} and W_{Ch} are fixed, while W_{rdt} and W_{PF} are trained).
- **Input Plastic:** Plasticity is strictly limited to its input synapses (W_{hh} and W_{MF} are fixed, while W_{ih}, W_{rdt}, W_{PF} and W_{Ch} are trained).
- **Fully Plastic:** All weights are trained.

Evaluation

The implementation will first be evaluated on standard reaching tasks simulated in MuJoCo, such as the Center-Out (CO) and Random Target (RT) tasks, to assess learning performance.



Center-out reach



Random target

Environment

The model to be used will be the tendon arm model from the MuJoCo physics engine, which is a widely used platform for simulating robotic and biomechanical systems with 6 controllable tendons.

